Ishaan Patel

ishbhagat02@gmail.com | linkedin.com/in/ishaan1402 | ishaan1402.github.io | github.com/ishaan1402

EDUCATION

NORTH CAROLINA STATE UNIVERSITY • M.S. in Computer Science, ML/AI Concentration Aug 2025 – May 2027
GPA: 3.78

UNIVERSITY OF NORTH CAROLINA AT CHARLOTTE • B.S. in Computer Science Aug 2020 – May 2024
GPA: 3.85 | Magna Cum Laude | Chancellor's List

EXPERIENCE

QUAZAR • Software Engineer Oct 2026 – Present

- Built a Python error monitoring service for a client's B2B SaaS platform, polling SQL Server logs and grouping errors into prioritized incidents.
- Integrated LLM-generated cause-and-fix summaries with alerting, redacting sensitive data before model calls.
- Designing an agentic workflow that traces urgent incidents into the codebase and opens pull requests with proposed fixes.

PROMETHEUS GROUP • Software Engineer Aug 2024 – Jun 2025

- Delivered 15+ features and resolved 40+ defects across the flagship product in C# .NET and React; refactored legacy components using Domain-Driven Design and planned sprint work for new customer contracts.
- Architected the flagship product's automated testing framework, the first in a company-wide QA initiative; triggered 100+ API and 30+ UI tests on every pull request in GitLab CI/CD and cut regression testing time by 40%.
- Mentored incoming developers through pair programming and code review; the framework became the standard for automated testing, adopted by 3+ product teams.

FIDELITY INVESTMENTS • Software Engineer Intern Jun 2023 – Aug 2023

- Developed a Java Spring Boot service for 401(k) transaction processing using a layered DAO/service architecture; configured Jenkins CI/CD and migrated the codebase from Bitbucket to GitHub.
- Implemented a message routing pipeline on Azure Service Bus with dead-letter queues to capture and retry failed messages.

UNIVERSITY OF NORTH CAROLINA AT CHARLOTTE • Undergraduate Researcher Jan 2023 – Jan 2024

- Adapted Microsoft's ALEX learned index to dynamic sparse graphs in C++ using 64-bit edge keys; benchmarked it against an STX B+ tree and CSR, VCSR, PCSR, and BAL formats on 10 SNAP graphs, with 1.5–2.8x faster edge insertion than the B+ tree.
- Parallelized PageRank, SSSP, BFS, and 1-hop workloads with OpenMP across 1–32 threads, measuring up to 20x speedup.

FIDELITY INVESTMENTS • Software Engineer Intern Jun 2022 – Aug 2022

- Built REST APIs and Java AWS Lambda functions for the Fidelity Bloom debit card roundup feature, monitored using CloudWatch.
- Deployed AWS CloudFormation stacks (S3, SQS, Lambda, KMS) to dev/QA environments through Jenkins pipelines.

PROJECTS

Upwind – Real-Time Air Quality Intelligence getupwind.me | github.com/Ishaan1402/upwind

- Built and deployed a live app that explains why local air quality spikes, pulling 11 sources such as AirNow, NASA FIRMS fire hotspots, satellite aerosol data, and weather to rank the most likely causes.
- Developed an LLM judge that verifies each explanation against the source data, swapping in a deterministic fallback when one fails, and runs nightly evals in GitHub Actions to fail the build when quality drops.

Eir – AI Triage for Ambulance Handoffs github.com/Ishaan1402/Eir

- Led a five-person team at the SAS Hackathon 2026 to build a live ambulance-to-hospital handoff app; it transcribes the crew's spoken report, extracts vitals tied to the exact words they came from, and suggests an ESI triage level.
- Benchmarked it on 1,683 real ambulance arrivals from CDC data; Eir placed 110 of 155 critical patients in the top two triage levels and flagged 38 more for clinician review.

crack-seg – UAV Bridge Crack Segmentation Live Demo | github.com/Ishaan1402/crack-seg

- Trained a 7.87M-param U-Net for pixel-level UAV bridge crack segmentation, achieving 0.730 Dice and 0.575 IoU on a held-out 447-image multi-source test set; presented at the 2026 NCSU AI Student Symposium.
- Deployed a FastAPI + Docker service to Hugging Face that handles 4K/8K UAV imagery via 448-px sliding-window inference.

SKILLS

- **Languages:** Python, C++, Java, C#, C, JavaScript
- **ML & AI:** PyTorch, LangGraph, Scikit-learn, Pandas, NumPy, Optuna
- **Cloud & DevOps:** AWS, Azure, Docker, GitHub Actions, GitLab CI/CD, Linux
- **Frameworks & Tools:** FastAPI, FastMCP, .NET, React, Spring Boot, PostgreSQL, Git